

Linear Maps, Change of Basis, Similarity

Linear maps and matrices

Here we stick to finite dimensional vector spaces.

Let $L : \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a linear map and let $E = \{e_1, \dots, e_n\}$ be any basis for $\mathbf{R}^n$ and $F = \{f_1, \dots, f_m\}$ be any basis of $\mathbf{R}^m$.

If $x = \sum x_i e_i$ is an element of $\mathbf{R}^n$, then $L(x)$ is an element of $\mathbf{R}^m$ and so there are unique scalars $y_1, \dots, y_m$ so that

$$L(x) = \sum_{i=1}^m y_i f_i.$$

If we represent x by the column vector

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_E,$$

where the subscript E means that the coordinates depend on the basis E , then

$$L(x) = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}_F$$

Linearity tells us that L is determined by its values $L(e_i)$ since

$$L(x) = \sum x_i L(e_i).$$

Define a matrix $[L]_E^F$ whose columns are the vectors

$$L(e_i) = \begin{bmatrix} y_{1i} \\ \vdots \\ y_{mi} \end{bmatrix}.$$

This yields an $m \times n$ matrix $[L]_E^F$ and by its construction it satisfies

$$[L(x)]_F = [L]_E^F [x]_E$$

This matrix $[L]$ is called the matrix of the linear map relative to the chosen bases.

Change of basis

The matrix $[L]_E^F$ depends on the bases E and F . Suppose we used different bases? For example, let E' and F' be different bases for $\mathbf{R}^n$ and $\mathbf{R}^m$.

How is the matrix $[L]_{E'}^{F'}$ related to $[L]_E^F$?

We can write the elements of E in terms of E' :

$$e_j = \sum_{i=1}^n a_{ij} e'_i$$

The columns of the matrix $A = (a_{ij})$ give the coordinates for the basis elements e_i in terms of the e'_i . Consequently, if $x = \sum b_i e_i$, then Ax gives the coordinates of x in the basis e'_i :

$$A[x]_E = A \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = [x]_{E'}.$$

Formally speaking, $A = [I]_E^{E'}$ is the matrix of the identity map where the “input” basis is E and the output basis is E' .

We can make a similar construction with the bases F and F' .

Then

$$[L]_E^F = [I]_{F'}^F [L]_{E'}^{F'} [I]_E^{E'}.$$

It's easy to see that $[I]_E^{E'} [I]_{E'}^E = [I]_E^E$, and this last matrix is the “identity” matrix with ones on the diagonal and zeros elsewhere. So

$$[I]_E^{E'} = ([I]_{E'}^E)^{-1}$$

Similarity

If $L : \mathbf{R}^n \rightarrow \mathbf{R}^n$, and E is any basis of $\mathbf{R}^n$, then $[L]_E^E$ is a square matrix. If E' is any other basis then

$$[L]_E^E = [I]_{E'}^E [L]_{E'}^{E'} [I]_E^{E'} = PAP^{-1}$$

where $P = [I]_{E'}^E$ and $A = [L]_{E'}^{E'}$. When two square matrices A and B are related by

$$A = PBP^{-1}$$

where P is an invertible square matrix, we say that A and B are similar.

In particular, all of the matrices $[L]_E^E$ associated to a single linear map in different bases are similar.

Proposition: Similar matrices have the same rank and nullity.

Let $e_1, \dots, e_k$ be a basis for the null space of A . If $B = PAP^{-1}$, then $Pe_1, \dots, Pe_k$ are linearly independent and belong to the null space of B , so the nullity of B is at least that of A . On the other hand, $A = P^{-1}BP$, so the same argument in reverse shows that the nullity of A is at least that of B , hence they are equal.

The equality of ranks follows since rank plus nullity is the number of columns.

In particular, the rank and nullity are well defined invariants of linear maps.

Isomorphisms

A linear map $L : V \rightarrow W$ is *injective* (or one-to-one) if and only if its nullity is zero. If the nullity is not zero, then there is a non-zero x with $L(x) = 0$, so L is not injective. On the other hand if $L(x) = L(y)$ but $x \neq y$, then $L(x - y) = 0$ but $x - y \neq 0$, so the nullity of L is greater than zero.

A linear map $L : V \rightarrow W$ is *surjective* (or onto) if and only if its rank equals the dimension of W . This is because the rank of L is the dimension of its image. So the image of L is a subspace of W whose dimension is the rank of L . This can only be all of W if the rank of L is the dimension of W .

An *isomorphism* between two vector spaces V and W is a linear map $L : V \rightarrow W$ with a linear inverse $G : W \rightarrow V$. It's enough for L to be bijective, meaning both injective and surjective, which forces V and W to have the same dimension.

Proposition: A linear map $L : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is invertible if and only if its nullity is zero.

An $n \times n$ matrix A is invertible if and only if it has rank n , or, equivalently, if it has nullity zero.